

$$\int \frac{(1+x)^2}{x(1+x^2)} dx = \int \frac{1+x^2+2x}{x(1+x^2)} dx = \int \frac{1+x^2}{x(1+x^2)} dx + \int \frac{2x}{x(1+x^2)} dx =$$

$$= \int \frac{1}{x} dx + 2 \int \frac{1}{1+x^2} dx = \ln x + 2 \arctan x + C$$

$$\int x \sin x dx = \left\{ \begin{array}{l} f = x \quad f' = 1 \\ g' = \sin x \quad g = -\cos x \end{array} \right\} = -x \cos x + \int \cos x dx =$$

$$= -x \cos x + \sin x$$

$$\int \frac{1}{1+9x^2} dx = \left\{ \begin{array}{l} t = 3x \\ dt = 3dx \\ \frac{1}{1+9x^2} dx = \frac{1}{1+t^2} \cdot \frac{1}{3} dt \end{array} \right\} =$$

$$= \int \frac{1}{1+t^2} \frac{1}{3} dt = \frac{1}{3} \cdot \int \frac{1}{1+t^2} dt = \frac{1}{3} \cdot \arctan t = \frac{1}{3} \arctan 3x + C$$

$$\int x \sqrt{x^2+4} dx = \left\{ \begin{array}{l} t = \sqrt{x^2+4} \\ dt = \frac{1}{2\sqrt{x^2+4}} \cdot 2x dx \\ dx = \frac{\sqrt{x^2+4}}{x} dt \\ x \sqrt{x^2+4} dx = x \cdot t \cdot \frac{\sqrt{x^2+4}}{x} dt = t^2 dt \end{array} \right\} =$$

$$= \int t^2 dt = \frac{1}{3} t^3 = \frac{1}{3} (\sqrt{x^2+4})^3 + C$$

$$\int \frac{\cos x}{1+\sin^2 x} dx = \left\{ \begin{array}{l} \cos x = t \\ -\sin x dx = dt \\ \frac{\cos x}{1+\sin^2 x} dx = \frac{t}{1+t^2} \cdot \frac{-dt}{\sin x} \end{array} \right\}$$

↑
będzie problematyczne,
więc nie podstawiamy

$$\rightarrow = \left\{ \begin{array}{l} \sin x = t \\ \cos x dx = dt \\ \frac{\cos x}{1+\sin^2 x} dx = \frac{\frac{dt}{dx}}{1+t^2} \cdot dx = \frac{1}{1+t^2} dt \end{array} \right\} =$$

$$= \int \frac{1}{1+t^2} dt = \arctg t = \arctg(\sin x) + C$$

$$\int \frac{x}{1+x^4} dx = \left\{ \begin{array}{l} x^2 = t \\ 2x dx = dt \\ \frac{x}{1+x^4} dx = \frac{dt}{2x} \cdot \frac{x}{1+t^2} = \frac{1}{2} \cdot \frac{1}{1+t^2} dt \end{array} \right\} =$$

$$= \int \frac{1}{2} \cdot \frac{1}{1+t^2} dt = \frac{1}{2} \int \frac{1}{1+t^2} dt = \frac{1}{2} \arctg t = \frac{1}{2} \arctg x^2 + C$$

$$\int \frac{1}{x(3\ln x + 5)} dx = \left\{ \begin{array}{l} t = 3\ln x + 5 \\ dt = 3 \cdot \frac{1}{x} dx \Rightarrow dx = \frac{dt \cdot x}{3} \\ \frac{1}{x(3\ln x + 5)} dx = \frac{1}{x \cdot t} \cdot \frac{dt \cdot x}{3} = \frac{1}{3} \cdot \frac{1}{t} \cdot dt \end{array} \right\} =$$

$$= \frac{1}{3} \int \frac{1}{t} dt = \frac{1}{3} \ln t = \frac{1}{3} \ln(3\ln x + 5) + C$$

Całkowanie funkcji wymiernych

$$\frac{W_l(x)}{W_m(x)} \quad \int \frac{W_l(x)}{W_m(x)} dx = ?$$

Na utanki proste można rozłożyć tylko

takie funkcje wymierne $f(x) = \frac{W_l(x)}{W_m(x)}$, gdzie $l < m$ jeśli

$l \geq m$ należy najpierw podzielić $W_l(x)$ przez $W_m(x)$ i zauważyć, że

$$\frac{W_l(x)}{W_m(x)} = \text{otrzymamy (iloraz)} + \frac{\text{Reszta}(x)}{W_m(x)}$$

$$\frac{x^3+1}{x^2+2} = x + \frac{1-2x}{x^2+2}$$

$$\begin{array}{r} x \\ x^3+1 : x^2+2 \\ -x^3-2x \\ \hline -2x+1 \end{array}$$

$$l < m$$

$$\frac{W_l(x)}{W_m(x)} = \frac{W_l(x)}{(x-a)(x-b)^k(x^2+p_1x+q_1)^{n_1} \dots} =$$

$\Delta < 0$ $\Delta < 0$

rozkładam
na dwa
utanki
proste

$$= \frac{A}{x-a} + \frac{B_1}{x-b} + \frac{B_2}{(x-b)^2} + \dots + \frac{B_k}{(x-b)^k} + \frac{P_1x+Q_1}{x^2+p_1x+q_1} + \frac{P_1x+Q_1}{x^2+p_1x+q_1} + \dots + \frac{\tilde{P}_1x+\tilde{Q}_1}{(x^2+p_1x+q_1)^2} + \dots + \frac{\tilde{P}_1x+\tilde{Q}_1}{(x^2+p_1x+q_1)^{n_1}}$$

$$\begin{aligned}\frac{x-2}{(x-1)(x+2)^2} &= \frac{a}{x-1} + \frac{b}{x+2} + \frac{c}{(x+2)^2} = \\ &= \frac{a(x+2)^2 + b(x+2)(x-1) + c(x-1)}{(x-1)(x+2)^2} = \\ &= \frac{ax^2 + 4ax + 4a + bx^2 + bx - 2b + cx - c}{(x-1)(x+2)^2} = \\ &= \frac{x^2(a+b) + (4a+b+c)x + 4a-2b-c}{(x-1)(x+2)^2}\end{aligned}$$

$$\begin{cases} a = -b \\ 4a + b + c = 1 \\ 4a - 2b - c = -2 \end{cases}$$

$$\begin{cases} -3b + c = 1 \Leftrightarrow -3b = 1 - c \\ -6b - c = -2 \quad -3b = 1 - \frac{4}{3} \end{cases}$$

$$2(1-c) - c = -2 \quad -3b = -\frac{4}{3}$$

$$2 - 3c = -2 \quad b = \frac{4}{9}$$

$$4 = 3c \quad a = -\frac{4}{9}$$

$$c = \frac{4}{3}$$

$$\int \frac{x^4}{(x^2-1)(x+2)} dx$$

$$(x^2-1)(x+2) = x^3 + 2x^2 - x - 2$$

$$\frac{x^4}{x^3 + 2x^2 - x - 2} = x + \frac{-x^4 - 2x^3 + x^2 + 2x}{x^3 + 2x^2 - x - 2}$$

$$\frac{x^4}{(x^2-1)(x+2)} = x + \frac{-2x^3 + x^2 + 2x}{x^3 + 2x^2 - x - 2}$$

$$\frac{-2}{x^3 + 2x^2 - x - 2} = \frac{-2}{2x^3 + 4x^2 - 2x - 4} = \frac{-2}{5x^2 - 4}$$

$$= x - 2 + \frac{5x^2 - 4}{(x^2-1)(x+2)}$$

$$\int \left(x - 2 + \frac{5x^2 - 4}{(x^2-1)(x+2)} \right) dx =$$

$$= \int x dx - 2 \int dx + \int \frac{5x^2 - 4}{(x^2-1)(x+2)} dx =$$

$$= \frac{1}{2}x^2 - 2x + \int \frac{5x^2 - 4}{(x^2-1)(x+2)} dx =$$

$$= \frac{1}{2}x^2 - 2x + \int \left(\frac{\frac{1}{6}}{x-1} + \frac{-\frac{1}{2}}{x+1} + \frac{\frac{16}{3}}{x+2} \right) dx =$$

$$= \frac{1}{2}x^2 - 2x + \frac{1}{6} \ln|x-1| - \frac{1}{2} \ln|x+1| + \frac{16}{3} \ln|x+2| + C$$

$$\begin{aligned}
 \frac{5x^2-4}{(x-1)(x+1)(x+2)} &= \frac{a}{x-1} + \frac{b}{x+1} + \frac{c}{x+2} = \frac{a(x+1)(x+2) + b(x-1)(x+2) + c(x^2-1)}{(x^2-1)(x+2)} = \\
 &= \frac{ax^2 + 3ax + 2a + bx^2 + bx - 2b + cx^2 - c}{(x^2-1)(x+2)} = \\
 &= \frac{x^2(a+b+c) + (3a+b)x + 2a-2b-c}{(x^2-1)(x+2)}
 \end{aligned}$$

$$\begin{cases}
 a+b+c=5 \\
 b=-3a \\
 2a-2b-c=-4
 \end{cases}$$

$$\begin{cases}
 -2a+c=5 \\
 2a+6a-c=-4
 \end{cases}$$

$$\begin{cases}
 c=5+2a \\
 8a-c=-4
 \end{cases}$$

$$8a - 5 - 2a = -4$$

$$6a = 1$$

$$a = \frac{1}{6} \quad c = \frac{32}{6} = \frac{16}{3}$$

$$b = -\frac{1}{2}$$